

Disparities in the prevalence, pathogenesis and progression of monoclonal gammopathy of undetermined significance and multiple myeloma between blacks and whites

.

There is marked racial disparity in the incidence of monoclonal gammopathy of undetermined significance (MGUS) and multiple myeloma, with a two to threefold increased risk in blacks compared with whites. The increased risk has been seen both in Africans and African Americans. Similarly, an increased risk of monoclonal gammopathies in blacks compared with whites has been noted after adjusting for socioeconomic and other risk factors, suggesting a genetic predisposition. The higher risk of multiple myeloma in blacks is likely a result of the higher prevalence of the premalignant MGUS stage; there are no data to suggest that blacks have a higher progression rate of MGUS to myeloma. Studies are emerging that suggest the baseline cytogenetic characteristics, and progression may differ by race. In contrast, to the increased risk noted in blacks, studies suggest that the risk may be lower in certain racial and ethnic groups, notably persons from Japan and Mexico. We review the literature on racial disparity in the prevalence, pathogenesis and progression of MGUS and multiple myeloma between blacks and whites. We also discuss future directions for research that could inform management of these conditions and positively influence patient outcomes.